

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · 4 product categories

DATA SCIENCE PROJECT



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating only

\$59.76

Avg Purchase

Demographics · Purchase details · Shopping behavior · Subscription status

Data Preparation in Python

Clean

Imputed missing Review Ratings by category median

Standardize

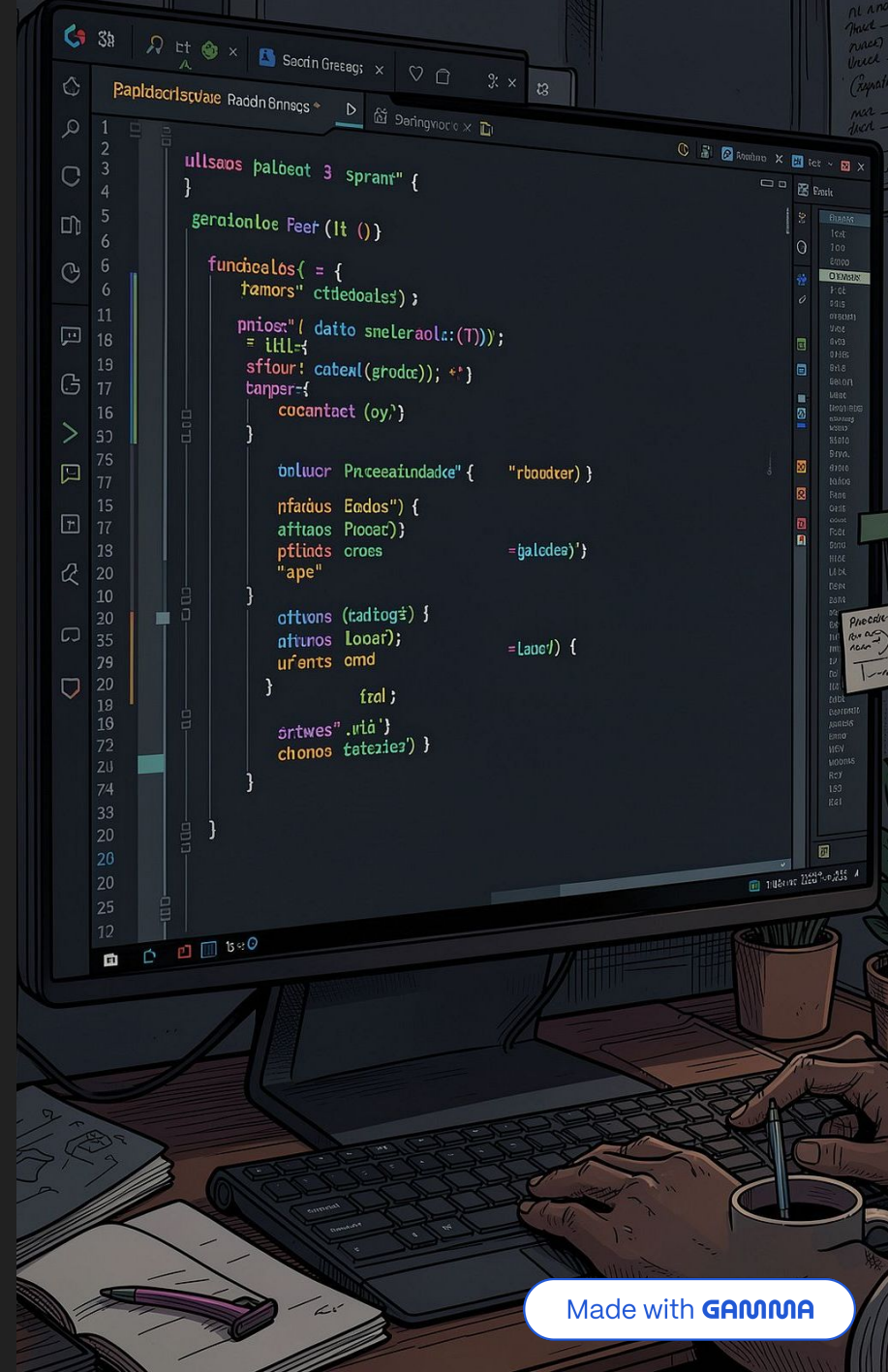
Renamed columns to snake_case

Engineer

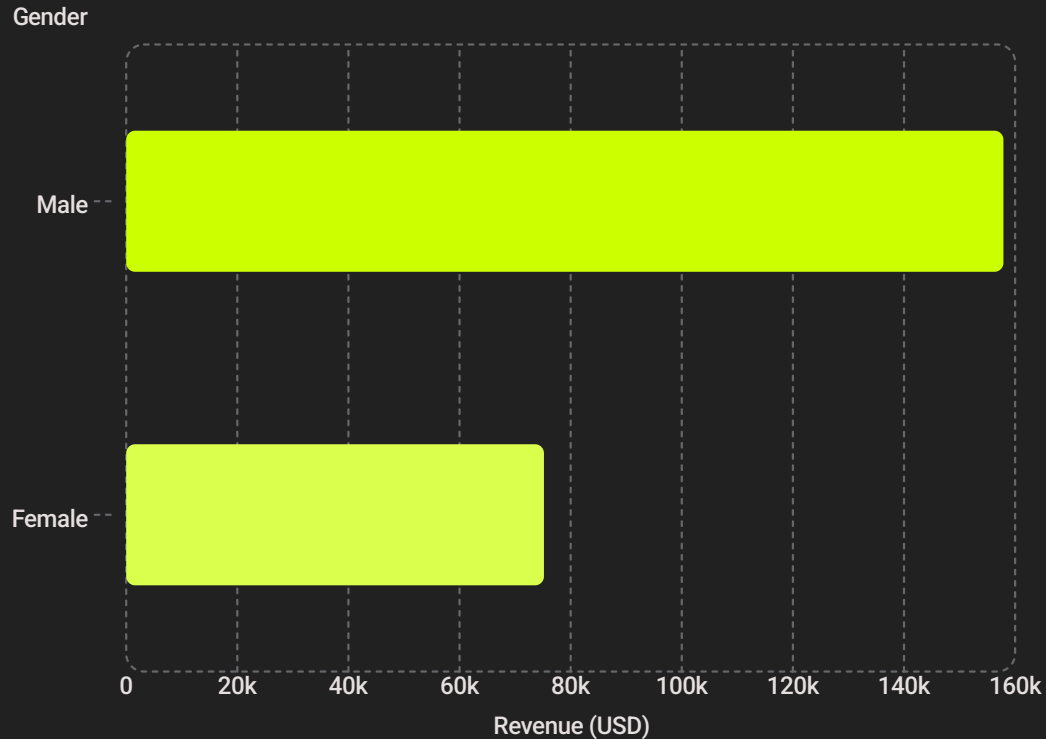
Created age_group & purchase_frequency_days

Load

Exported cleaned data to PostgreSQL



Revenue by Gender



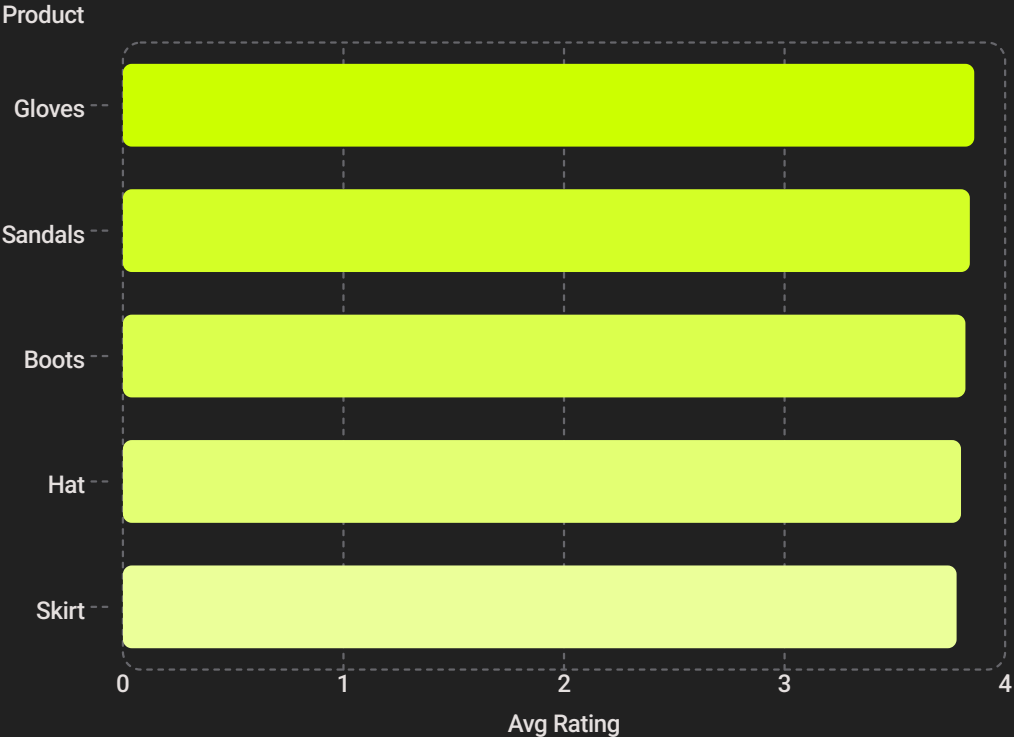
Key Finding

Male customers generated **2.1× more revenue** than female customers — \$157,890 vs \$75,191.

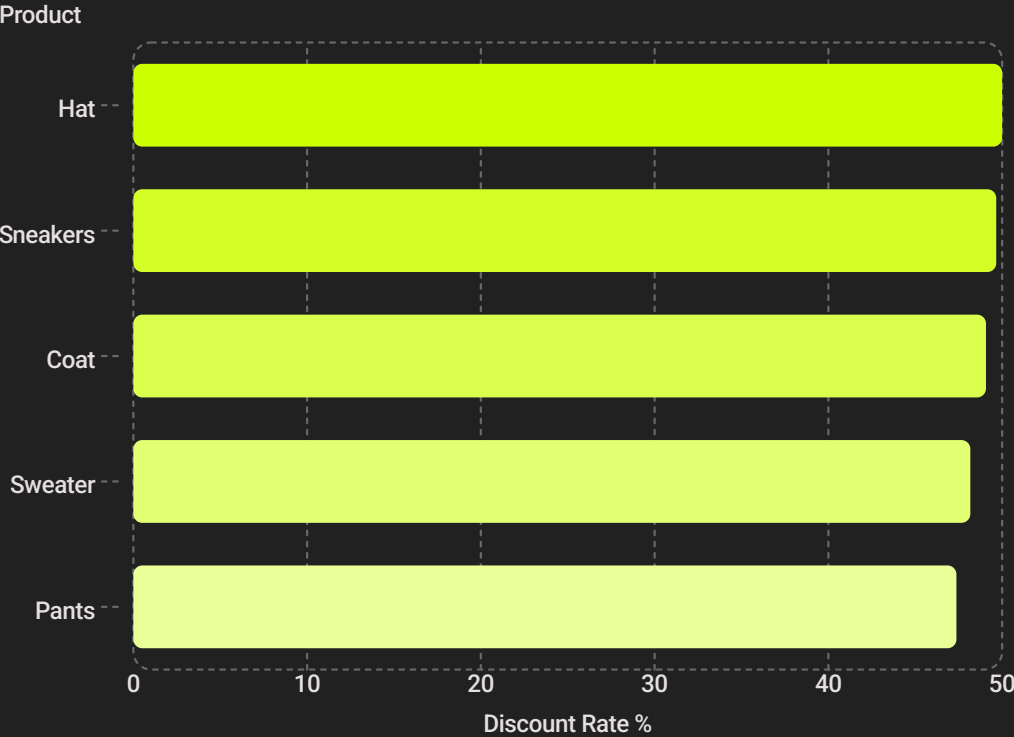
- ❑ Male shoppers represent 68% of all customers (2,652 of 3,900).

Top Products & Discount Patterns

Highest-Rated Products

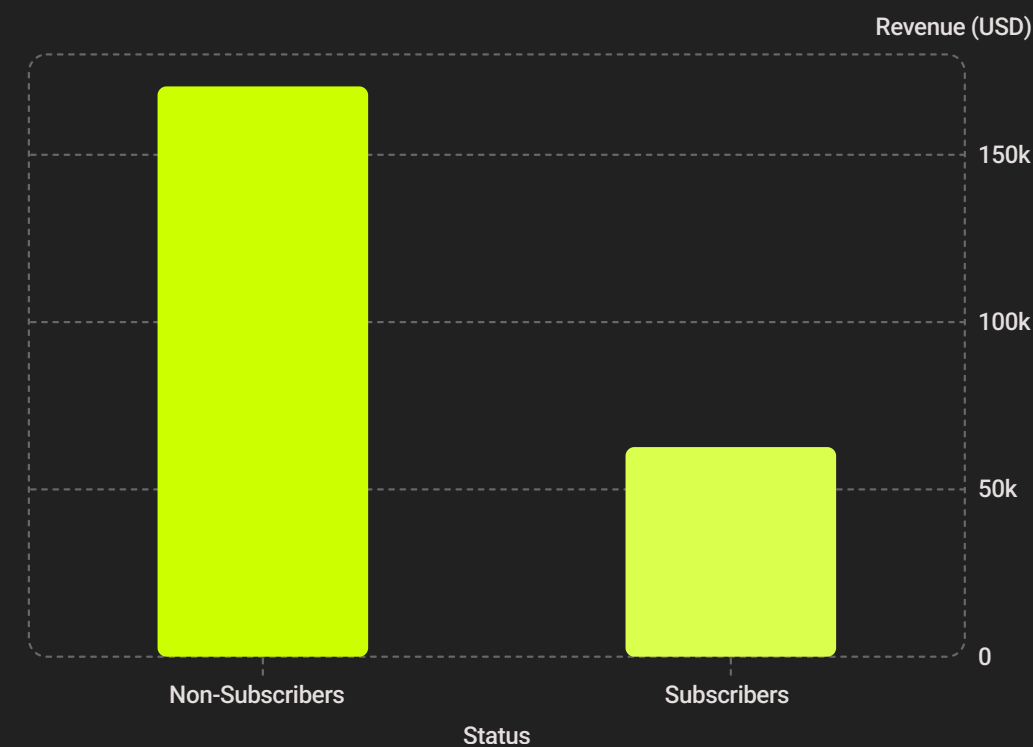


Most Discounted Products

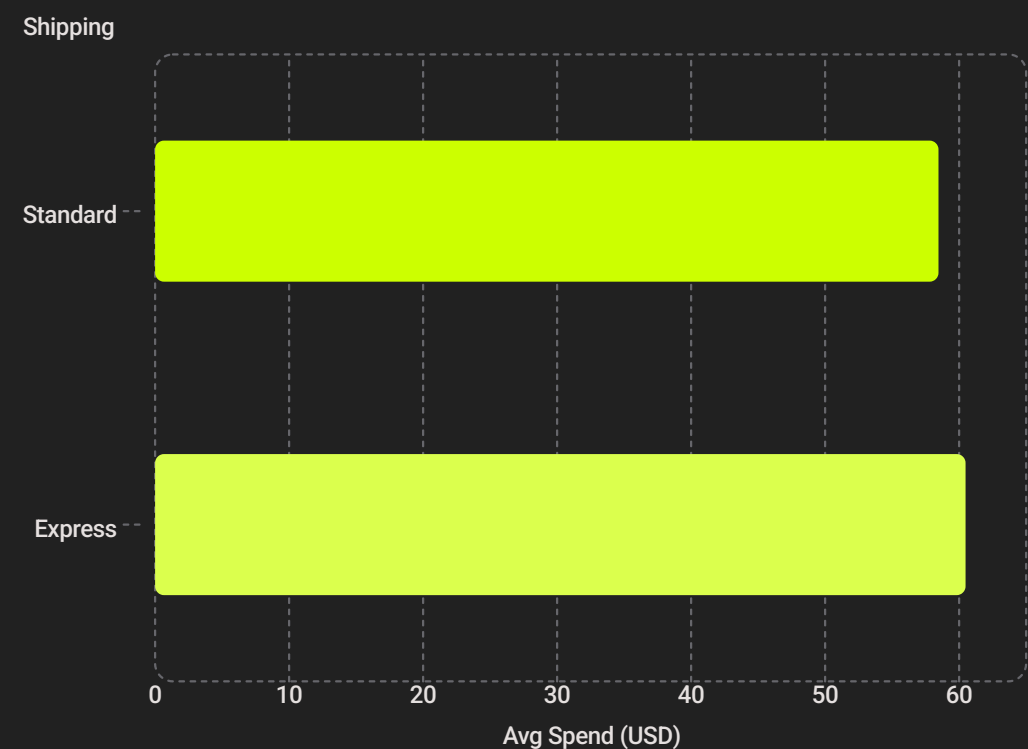


Subscription & Shipping Insights

Subscribers vs Non-Subscribers

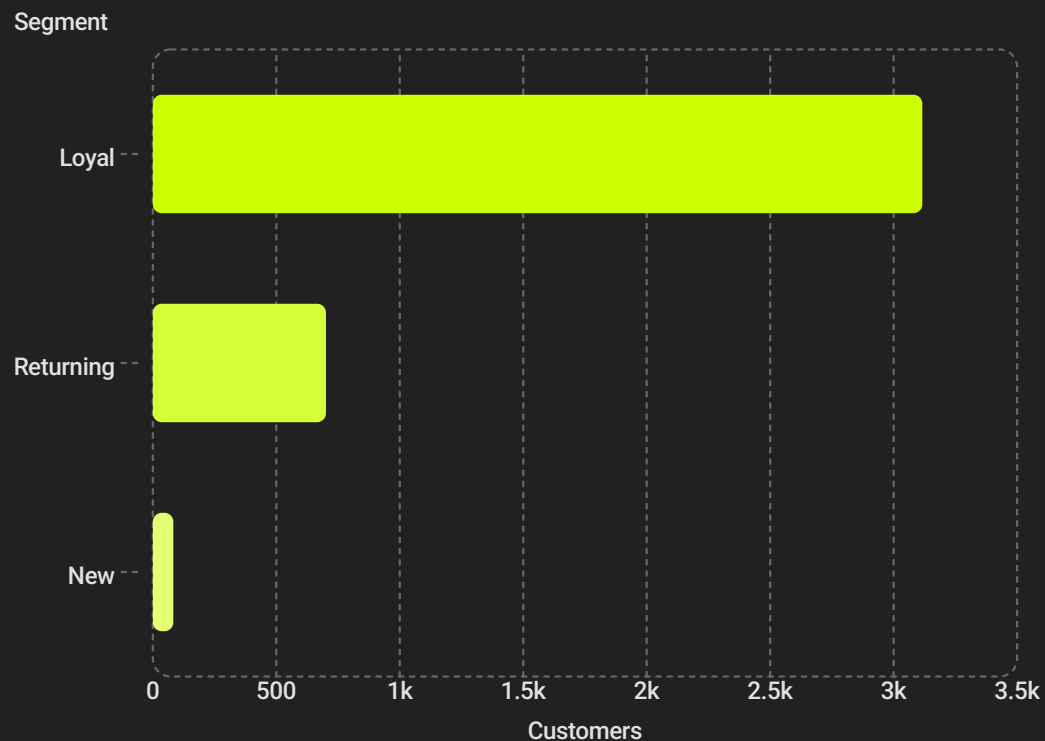


Shipping Type Avg Spend



Express shippers spend slightly more — a **targetable segment** for upsell campaigns.

Customer Segmentation



Segment Breakdown

Loyal – 80%

3,116 customers

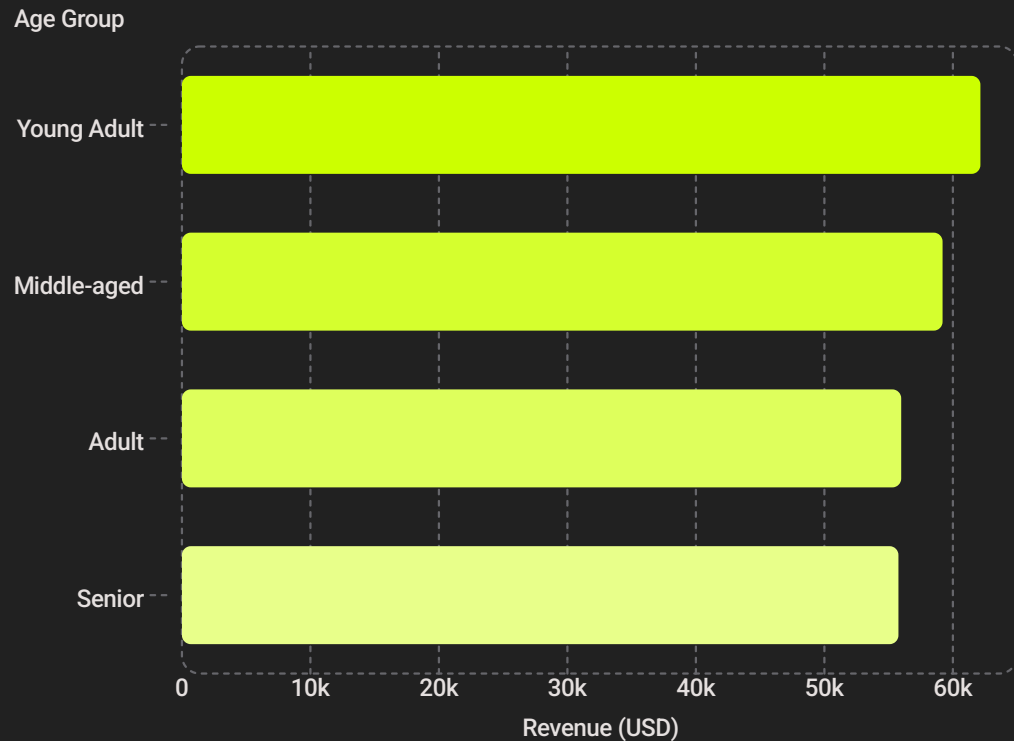
Returning – 18%

701 customers

New – 2%

83 customers

Revenue by Age Group



Age Group Insights

Young Adults lead in revenue at **\$62,143**, but all four age groups contribute relatively evenly — no single group dominates.

- Repeat buyers with 5+ purchases: 958 subscribers vs 2,518 non-subscribers.



Power BI Dashboard

Key Metrics

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Revenue by Category

Clothing \$100K · Accessories
\$75K · Footwear \$30K ·
Outerwear \$15K

Subscription Split

27% subscribers · 73% non-subscribers



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers